

INDUSTRIAL INTERNSHIP REPORT

**PROFESSIONAL CERTIFICATION IN MICROSOFT
AZURE AI FUNDAMENTALS (AI-900)**

Submitted in Partial Fulfillment of the Requirements for the Degree of

**Bachelor of Engineering
In**

**COMPUTER SCIENCE ENGINEERING (ARTIFICIAL
INTELLIGENCE AND MACHINE LEARNING)**

by

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This is to certify that the industrial internship report entitled “**PROFESSIONAL CERTIFICATION IN MICROSOFT AZURE AI FUNDAMENTALS (AI-900)**” submitted by “Armaan Atri” in partial fulfillment of the requirements for the award of the **Degree Bachelor of Engineering** in “**COMPUTER SCIENCE ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)**” is a bonafide record of the work carried out under my(our) guidance and supervision at Chandigarh University, Gharuan.

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Program Leader - **AI-ML**

Apex Institute of Technology

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Declaration by Candidate

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ABSTRACT

This industrial internship report presents the learning outcomes and practical exposure gained through the completion of the Microsoft Azure AI Fundamentals (AI-900) Professional Certification. The primary objective of this internship was to understand the foundational concepts of Artificial Intelligence (AI) and explore their real-world implementation using cloud-based services provided by Microsoft Azure.

The problem addressed in this training was the lack of structured, industry-oriented exposure to modern AI technologies, including Machine Learning, Generative AI, Natural Language Processing (NLP), Speech AI, Computer Vision, and Information Extraction. With the rapid growth of AI-driven applications, it has become essential for engineering students to acquire hands-on experience with scalable, cloud-powered AI platforms.

The approach followed in this internship combined theoretical learning modules with practical exercises in Azure Machine Learning Studio, Azure Cognitive Services, Azure AI Foundry, and AI Playground. Each module included guided tutorials, sample datasets, interactive labs, and real-time experimentation using vision, speech, text, and document intelligence tools.

The major results of this internship include a strong conceptual understanding of AI and ML workflows, familiarity with cloud-based AI services, practical experience in sentiment analysis, speech-to-text conversion, image recognition, OCR, and document processing. The certification strengthened problem-solving abilities and improved technical competency in designing, evaluating, and understanding AI solutions.

In conclusion, this internship successfully enhanced both theoretical and practical knowledge in Artificial Intelligence, enabling the learner to understand the complete lifecycle of AI development using Azure services. The skills gained through this certification contribute significantly toward industry readiness and future specialization in AI and Machine Learning.

Keywords: Artificial Intelligence, Machine Learning, Azure Cognitive Services, NLP, Computer Vision, Speech AI, Generative AI, Document Intelligence, Cloud Computing.

NOMENCLATURE

Term / Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
LLM	Large Language Model
NLP	Natural Language Processing
OCR	Optical Character Recognition
API	Application Programming Interface
GPU	Graphics Processing Unit
SDK	Software Development Kit
JSON	JavaScript Object Notation
Azure ML	Azure Machine Learning
OCR	Optical Character Recognition
CV	Computer Vision
TTS	Text-to-Speech
STT	Speech-to-Text
NER	Named Entity Recognition
KPI	Key Performance Indicator
UI	User Interface
CPU	Central Processing Unit
SDK	Software Development Kit

CHAPTER - 1

1. Introduction

This industrial internship report is based on the completion of a professional certification in **Microsoft Azure AI Fundamentals (AI-900)**. The training focused on building foundational knowledge of Artificial Intelligence, Machine Learning, Generative AI, Natural Language Processing, Speech AI, Computer Vision, and AI-powered information extraction using Azure tools. The certification provided both theoretical understanding and practical exposure to cloud-based AI solutions through guided modules and hands-on labs offered in Microsoft Learn and Azure AI Foundry.

1.1 Topic addressed

The internship addresses the topic **“Foundational Concepts and Practical Implementation of Artificial Intelligence using Microsoft Azure AI Services.”**

The major areas of study include:

- Understanding core concepts of Artificial Intelligence
- Machine Learning fundamentals and workflows
- Generative AI and Large Language Models
- Natural Language Processing concepts and applications
- Speech recognition and synthesis
- Computer Vision fundamentals and Azure Vision services
- Information extraction using Azure Document Intelligence

This topic focuses on how AI models are trained, evaluated, deployed, and consumed through cloud-based solutions using Azure’s cognitive APIs.

1.1.1 Importance / Novelty of the Topic

The significance of this topic lies in its relevance to modern technological development. AI is reshaping industries through automation, intelligent decision-making, and data-driven insights. The novelty includes:

- Integration of AI with cloud platforms for scalable solutions
- Adoption of cognitive services for low-code/no-code AI development
- Use of Generative AI and LLMs in real-time content generation
- Ensuring Responsible AI practices such as fairness, transparency, and reliability
- Practical exposure to end-to-end AI workflows within Microsoft Azure

Understanding these areas is essential for building career-ready AI solutions aligned with industry requirements.

1.1.2 Related Literature / Previous Work

Artificial Intelligence and Machine Learning concepts have been studied and improved by various researchers and technology providers:

- It has been reported in multiple studies that cloud-based AI accelerates intelligent application development by providing pre-trained models and scalable infrastructure.
- Jiang et al. highlighted that machine learning involves data preprocessing, feature engineering, training, evaluation, and deployment workflows.
- Microsoft's Responsible AI guidelines provide insights on fairness, transparency, and governance in AI application development.
- Research shows that NLP, Speech AI, and Computer Vision are critical technologies behind chatbots, document processing, voice assistants, biometric systems, and automated image analysis.
- Azure documentation and industry case studies demonstrate real-world applications in healthcare, finance, manufacturing, retail, and enterprise automation.

CHAPTER - 2

2. Weekly Report

The approach used during the industrial internship was a structured and module-based learning methodology offered through the Microsoft Learn platform and Azure AI services. The certification training followed a systematic framework combining theoretical understanding, hands-on practice, guided exercises, and assessment-based learning. This approach ensured clarity of concepts, practical skill development, and achievement of measurable learning outcomes aligned with the goals of the AI-900 certification.

The industrial training program was divided into logical learning units that progressively introduced AI concepts, machine learning workflows, cloud-based AI implementations, generative AI capabilities, and cognitive services. Each module and activity was designed to build conceptual understanding first, followed by practical applications in the Azure environment. Weekly progression was tracked through learning XP, unit completion, practical exercises, and applied scenarios.

WEEKLY REPORT

WEEK 1 21/10/2025 to 25/10/2025	TOPIC COVERED
DAY-1 21/10/2025	Introduction to Machine Learning Concepts • Definition of AI • Types of AI systems • Real-world applications • Difference between narrow vs general AI • Ethical considerations
DAY-2 22/10/2025	Introduction to Machine Learning Concepts • Supervised, Unsupervised & Reinforcement Learning • Classification vs Regression • Training, Testing & Validation • Model accuracy & evaluation methods • Data requirements and preprocessing
DAY-3 23/10/2025	Get Started with Machine Learning in Azure • Designing ML training pipelines • Using Azure ML workspace • Understanding datasets, compute clusters, environments • Automating ML experiments
DAY-4 24/10/2025	Introduction to Generative AI & Agents • What is Generative AI? • Introduction to LLMs • Prompt engineering basics • Agent-based reasoning
DAY-5 25/10/2025	Get Started with Generative AI in Azure • Azure AI Foundry portal overview • Model marketplace usage • Testing models in playground • Understanding token limits, temperature, response behavior

WEEK 2 27/10/2025 to 31/10/2025	TOPIC COVERED
DAY-1 27/10/20	Introduction to Natural Language Processing Concepts • Introduction to NLP concepts • Tokenization, stemming, and lemmatization • Sentiment analysis • Key phrase extraction • Named Entity Recognition (NER) • Overview of Azure AI Language services
DAY-2 28/10/20	Computer Vision Basics • Introduction to computer vision concepts • Image classification • Object detection • Feature extraction • OCR (Optical Character Recognition) • Azure Vision capabilities
DAY-3 29/10/20	Introduction to AI Speech Concepts • Introduction to AI speech concepts • Speech-to-text processing • Text-to-speech synthesis • Acoustic and language modeling • Azure AI Speech Studio overview • Real-time transcription workflow
DAY-4 30/10/20	Azure Computer Vision Services • Azure Vision APIs • Face detection • Landmark detection • Text extraction from images • Confidence scores & bounding boxes • Image analysis in Azure portal
DAY-5 31/10/20	Information Extraction & Document Intelligence • Introduction to AI-powered information extraction • Layout and document structure analysis • Key-value pair extraction • Form and receipt analysis • Overview of Azure Document Intelligence Studio • Extracting structured fields from invoices and documents

2.2 Tools and Platforms Used

The following tools and platforms were used throughout the internship to complete training modules, perform hands-on tasks, and explore practical implementations of AI solutions:

- **Microsoft Learn Platform:** Used for accessing structured learning modules, theory content, guided exercises, and assessments.
- **Azure Portal:** Provided access to cloud services, dashboards, and configuration panels needed for AI services.
- **Azure Machine Learning Studio:** Used for designing and running ML training pipelines, automated ML experiments, and monitoring model performance.
- **Azure Cognitive Services:** Enabled practical implementation of Vision, Speech, Language, and Document Intelligence capabilities.
- **Azure AI Foundry:** Used for building, testing, and experimenting with generative AI models, LLM playground, and prompt engineering configurations.
- **Azure AI Playground:** Helped in testing cognitive APIs like OCR, sentiment analysis, face detection, speech-to-text, and document analysis.
- **Microsoft Certification Portal (Exam Prep):** Used for practice assessments, knowledge checks, and exam readiness evaluation.

These tools provided an enterprise-level technological environment, enabling simulation of real-world AI systems and practical application of AI concepts learned during the certification.

CHAPTER - 3

3. Conclusions and Recommendations

3.1 Conclusions

The industrial internship, undertaken through the Microsoft Azure AI Fundamentals (AI-900) Professional Certification, provided a strong foundation and a comprehensive understanding of Artificial Intelligence and its cloud-based implementation using Azure services. Over the span of two structured weeks, the training offered in-depth exposure to essential AI concepts, machine learning workflows, generative AI capabilities, natural language processing techniques, speech recognition, computer vision, and document intelligence.

The hands-on approach followed throughout the certification helped in bridging the gap between theoretical concepts and their real-world applications. By performing practical exercises using Azure Cognitive Services, Azure Machine Learning Studio, and Azure AI Foundry, it became clear how AI solutions are built, tested, deployed, and consumed within production environments.

Some key conclusions drawn from the internship include:

- The Azure ecosystem offers scalable, easy-to-use AI tools that significantly reduce the complexity of AI development.
- Understanding core concepts of AI, ML, and Generative AI forms the basis of building intelligent systems.
- Cognitive services allow quick implementation of NLP, Vision, Speech, and Document analysis without requiring deep ML expertise.
- Responsible AI practices are essential to ensure transparency, fairness, reliability, and ethical deployment of AI solutions.
- Real-world use cases such as sentiment analysis, speech recognition, image analysis, and document processing are easily achievable through Azure APIs.

Overall, the internship effectively enhanced theoretical knowledge, improved technical proficiency, and strengthened readiness for industry-level AI roles.

3.2 Recommendations

Although the internship successfully met its objectives, several improvements and future developments can further enhance learning outcomes:

1. **Increase Hands-On Project Work:** More real-world mini-projects such as chatbot creation, invoice processing automation, or image classification can strengthen applied skills.
2. **Work with Live Data:** Using real datasets instead of sample datasets will provide more realistic exposure to data cleaning, preprocessing, and performance evaluation.
3. **Explore Custom Model Building:** Students should explore training custom NLP or Vision models rather than relying only on pre-trained cognitive services.
4. **Extend Deployment Knowledge:** Practical exercises on container-based deployment, REST API integration, and endpoint security will deepen understanding of production-level AI systems.
5. **Try Multi-Service Integration:** Combining Vision + NLP + Speech services into a single intelligent application will improve problem-solving skills and system-design capabilities.
6. **Stay Updated with Azure Developments:** Azure AI services evolve rapidly. Students should periodically revisit learning modules to stay updated with new AI capabilities and features.
7. **Focus on Responsible AI Implementation:** Additional emphasis on fairness, bias detection, privacy, and transparency will ensure ethical AI development practices.

APPENDICES

APPENDIX A – Course Syllabus

Course Code: 23CSI-338

Course Title: Industrial Internship

Duration: Minimum 6 Weeks

Course Objectives:

1. To expose students to industry practices.
2. To enable students to apply technical knowledge in real-world settings.
3. To enhance analytical, technical, and communication skills.
4. To develop familiarity with industrial tools, workflows, and professional expectations.

Course Outcomes (COs):

- CO1: Demonstrate practical technical skills gained through internship/certification.
- CO2: Understand industry-oriented tools and methods.
- CO3: Apply AI/ML/Cloud concepts in real-world scenarios.
- CO4: Present internship findings professionally.

Requirements:

- Completion of internship OR professional certification
- Weekly report
- Certificate
- PPT presentation
- Viva voce

APPENDIX B – Certificate

Microsoft Certified

Azure AI Fundamentals

Armaan Atri

has successfully completed the requirements of
Azure AI Fundamentals

Date Issued: November 3, 2025



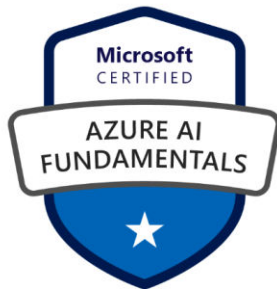
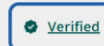
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Microsoft Certified: Azure AI Fundamentals

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Earners of the Azure AI Fundamentals certification have demonstrated foundational knowledge of machine learning (ML) and artificial intelligence (AI) concepts and related Microsoft Azure services.

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Skills

Azure Bot Services

Azure Machine Learning

Cognitive Services

Earning Criteria

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EXAM SCORE REPORT

CANDIDATE	EXAM
Armaan Atri 23bai70302@cuchd.in	AI-900: Microsoft Azure AI Fundamentals Registration ID: ITS-49216354 Exam reference #: 50616531 Date: November 3, 2025 ID: 23bai70302@cuchd.in

RESULTS	100	200	300	400	500	600	700	800	900	1000
Required Score										
Your Score										

SECTION ANALYSIS		FINAL SCORE	
Describe Artificial Intelligence Workloads and Considerations (15-20%)	100%	Required Score	700
Describe Fundamental Principles of Machine Learning on Azure (15-20%)	86%	Your Score	833
Describe Features of Computer Vision Workloads on Azure (15-20%)	80%	OUTCOME	
Describe Features of Natural Language Processing (NLP) Workloads on Azure (15-20%)	85%		
Describe features of generative AI workloads on Azure (20-25%)	75%		
		Pass	✓

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APPENDIX C – Fee Receipt

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	UNIVERSITY INSTITUTE OF ENGINEERING (UIE) Year/Semester 3/5
	Program : Bachelor of Engineering (Computer Science and Engineering) (Hons.) (with Specialization in Artificial Intelligence and Machine Learning) (In association with IBM)
Received With Thanks from Mr. / Ms. ARMAAN ATRI S/D/o Sh. AJAY ATRI	
On Accounts of Institutional Training:2245	
In form of Cash/Draft/Pay Order/On line (Specify) By Online No. (Serial No) UM3308473	
Drawn on HDFC-472646 Dated 11-06-2025 Payable At	
Rs. 2245/-	(In words) Rs. Rupees Two Thousand Two Hundred Forty Five ONLY ONLY
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7. Disputes, if any, shall be subject to the jurisdiction of the courts in District Mohali (Punjab) only.	
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